A011 — Maxwell SST Kinetic Falsifier
v0.4.1 Scientific Protocol and Falsifier Report

SST-Workbench

Framework v1.0.10; controlled geometry diagnostic tranche

1 Identification, version and authority

A011 v0.4.1 is a packaging/report binding patch of the preserved v0.4.0 geometry pilot. It uses the byte-qualified SST Falsifier Framework v1.0.10. Equations, source geometry, parity thresholds and physical input requirements are unchanged. Scientific claims remain unresolved; software execution is not physical certification.

2 Research question and scientific objective

Does the v1.0.10 adapter retain inherited A011 geometry definitions and Python/native consistency on exact admitted PKLSA inputs?

Qualify software migration and bounded geometry diagnostics while explicitly withholding physical closure.

3 Null and alternative hypotheses; falsification criteria

H_0. Adapter source admission or the inherited native/reference tolerances fail on the preregistered geometry set.

H_1. Exact admitted geometry is processed with finite inherited diagnostics and existing native/reference tolerances; no physical prediction follows.

- F1: Any source hash/capability mismatch aborts execution; no substitution. Recorded consequence: FAIL_INFRASTRUCTURE
- F2: Native parity outside original tolerances is a failed implementation smoke. Recorded consequence: FAIL
- F3: Unavailable independent physical inputs cannot support physical discovery or closure. Recorded consequence: NOT_RUN_PREREQUISITE

4 Assumptions, domain and edge cases

1. Input coordinate units are source-native and have no certified SI scale.
2. Closed polygon curves are resampled by the unchanged A011 reference definition.
3. Two resolutions expose sensitivity only.
4. No energy spectrum, gravity or thermodynamic state-count evidence is inferred from curves.

Registered edge cases: Missing native ABI-compatible binary; Raw/geometry hash drift; Nonfinite or degenerate geometry; Existing output directory; No calibrated SI scale

5 Symbols, definitions, SI units and dimensional audit

Symbol	Definition	SI unit / scale	Dimensions
p_i	Source-native polygon vertex	Uncalibrated source coordinate unit	L in a declared native unit
L	Resampled polygon length	Uncalibrated source coordinate unit	L
Wr	Inherited midpoint writhe diagnostic	1	1

$v_{cpp,j}$ and $v_{py,j}$ are the corresponding native/reference kernel-test components in E2; N counts resampled points. Curvature has inverse native-coordinate units. There is no independently calibrated SI length/time conversion. Gamma=1, smoothing=0.05 and the fixed probe offset are nondimensional software probes, not measured physical inputs.

6 Registered equations and code-to-mathematics mapping

Machine-readable science contract

Research question Does the v1.0.10 adapter retain inherited A011 geometry definitions and Python/native consistency on exact admitted PKLSA inputs?

Objective Qualify software migration and bounded geometry diagnostics while explicitly withholding physical closure.

H_0 Adapter source admission or the inherited native/reference tolerances fail on the preregistered geometry set.

H_1 Exact admitted geometry is processed with finite inherited diagnostics and existing native/reference tolerances; no physical prediction follows.

Registered equations

E1: Inherited polygon length

$$L = \sum_i \|p_{i+1} - p_i\| \tag{1}$$

Dimensional check: Coordinate length; no SI conversion without calibration.

E2: Existing componentwise velocity parity criterion

$$|v_{cpp,j} - v_{py,j}| \le 10^{-12} + 10^{-11}|v_{py,j}| \tag{2}$$

Dimensional check: Same nondimensional test-kernel convention on both implementations; no physical velocity claim.

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Rehash qualified source, protocol, exact admitted raw and parsed geometry inputs
2. **S2** [G2; equations E1]: Apply unchanged A011 resampling and geometry definitions
3. **S3** [G5; equations E2]: Compare inherited native and transparent reference kernels despite missing physical prerequisites

Only the equations registered in the inherited science contract are asserted here. Other inherited geometry diagnostics are defined by the cited implementation; no missing physical model is invented.

7 Algorithm, phases and execution order

1. Verify qualified framework, frozen geometry protocol and report/output binding.
2. Run targeted tests before any fresh bounded smoke.
3. Rehash the three admitted carrier descriptors, source bytes, parsed geometry and qualification evidence.
4. Apply the unchanged A011 resampling and geometry kernels at N=64 and N=128.
5. Attempt independent implemented evaluators despite unavailable physical prerequisites; record raw outcomes and recursive qualification separately.
6. Seal machine evidence; generate scientific TeX with the existing framework; compile the PDF; publish a unique ZIP and SHA-256 sidecar.

8 Sources, provenance and dataset qualification

Only the exact sources in source_contract.json and the Phase 2 A011 admission ledger are used. Raw and parsed geometry hashes and prerequisite capability records are reverified. All three are numerical KnotPlot geometry sources, not independent physical replications. The complete current source manifest is included by the automatic results section. The qualified framework and E010 parser pins remain separate from scientific qualification.

9 Migration, execution and scientific gates

Migration, execution and scientific statuses are separate axes. G0/G1 verify protocol/source integrity; G2 measures finite geometry diagnostics. G5 measures actual native/reference parity but remains DIAGNOSTIC_ONLY when G3/G4 physical prerequisites are unavailable. G3/G4/G7/G8 remain not executed for missing evidence. GPU G6 is disabled and reveal G9 is deferred. Exact current records, qualifications and allows-dependent-claim fields are generated directly from the machine ledger below.

10 Numerical methods, tolerances and convergence criteria

Unchanged uniform polygon arclength resampling, discrete geometry diagnostics and FP64 kernels from A011 v0.3.1 are used. The frozen ladder is $N = 64, 128$. Native/reference velocity uses rtol=10^{-11} and atol=10^{-12}; writhe and distance require absolute deltas $< 10^{-11}$. No NaN/Inf, source drift or missing native module is replaced by a warning. These two resolutions expose sensitivity only: no spatial, time, solver or regularization convergence certificate is claimed. Missing physical error budgets remain unresolved.

11 Backend manifest and authority

CPython 3.14/NumPy FP64 is the transparent reference. The inherited byte-pinned Windows cp314 C++17 kernel is used without rebuilding or silent fallback; its legacy backend version remains 0.3.1-cpp17. This is geometry-kernel implementation evidence only. No GPU, physical solver parity or independently authenticated compiler rebuild is claimed. Actual environment and binary/source hashes are rendered below.

12 Blind/reveal boundaries

No private target is opened, no expected winner is optimized and no reveal is performed. This is a nonrevealed geometry-diagnostic adapter run, not the complete private-target BLIND/REVEAL experiment. The generic framework automatic provenance uses BLIND for unrevealed output; in this report that label does not assert a completed physical blind campaign. No historical dataset seal or blind/reveal commitment is modified.

13 Uncertainty, independence and statistical limits

The three representative topologies are fixed in advance; carrier ordering is lexicographic after exact admission and size filters. There is no random sample, fitted parameter, p-value or population-level significance claim. The common source lineage does not provide independent replication. Resolution differences, parity residuals and missing physical uncertainty are reported separately.

14 Automatic results, tables and diagnostic graphs

Separate migration, execution and scientific axes

Axis	Actual status
migration_gate	PASS_GEOMETRY_ADAPTER
execution_gate	PASS_BOUNDED_GEOMETRY_SMOKE
scientific_gate	UNRESOLVED_MISSING_PHYSICAL_INPUTS
scientific_status_promoted	False

Measured geometry and resolution sensitivity

Topology	N	Native length	Mean curvature	Midpoint writhe
0_1	64	12.52869104	0.5033844161	0
0_1	128	12.54017859	0.6604430041	0
3_1	64	16.32543462	0.9829584087	3.412398906
3_1	128	16.36257488	1.034079532	3.408273635
4_1	64	22.35286777	0.910604241	0.2068828022
4_1	128	22.44280283	0.9558892656	0.2007498053

Two registered levels expose sensitivity; they do not establish convergence or physical calibration.

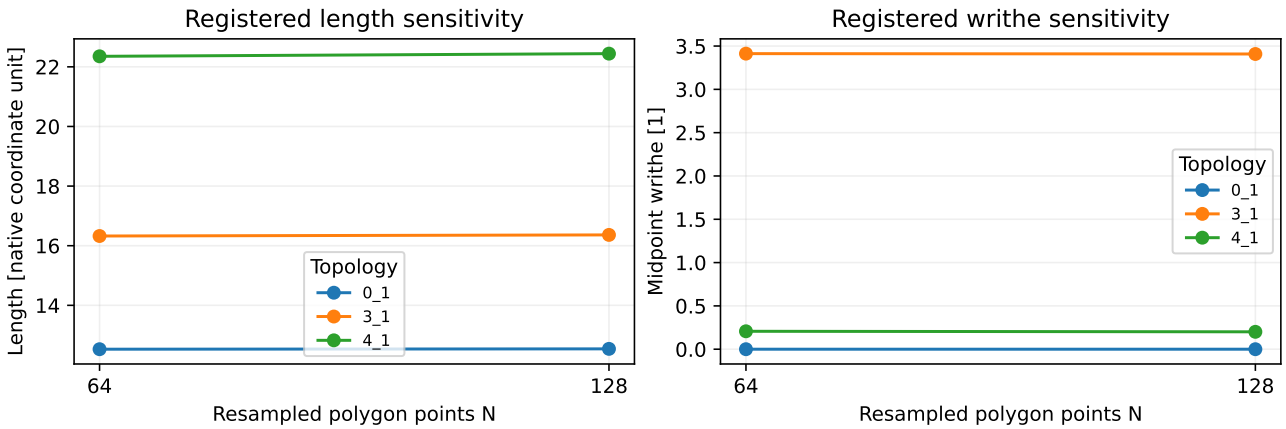


Figure 1: Actual N=64 and N=128 measurements from PLOT_DATA.json. Native lengths have no calibrated SI scale; writhe is dimensionless. No fitted trend or continuum extrapolation.

Actual native/reference comparisons

Carrier topology	N	Velocity max abs	Writhe abs	Distance abs
0_1	64	2.7756e-17	0	0
0_1	128	6.9389e-18	0	0

3_1	64	5.5511e-17	0	0
3_1	128	5.5511e-17	0	0
4_1	64	2.2204e-16	0	0
4_1	128	2.2204e-16	0	0

Exact gate execution and claim authority

Gate	Execution	Qualification	Allows dependent claim
G0	EXECUTED	QUALIFIED	True
G1	EXECUTED	QUALIFIED	True
G2	EXECUTED	QUALIFIED	True
G3	NOT_EXECUTED	BLOCKED_PREREQUISITE	False
G4	NOT_EXECUTED	BLOCKED_PREREQUISITE	False
G5	EXECUTED	DIAGNOSTIC_ONLY	False
G6	DISABLED	NOT_APPLICABLE	False
G7	NOT_EXECUTED	BLOCKED_PREREQUISITE	False
G8	NOT_EXECUTED	BLOCKED_PREREQUISITE	False
G9	DEFERRED	DEFERRED	False

Missing scientific evidence

- Calibrated physical length and SI source scale
- Independent physical energy spectrum/gaps and relaxation/couplings
- Independent thermodynamic sampled state counts
- Held-out force/pressure and mechanism validation
- Spatial/time/solver/regularization convergence beyond this two-level geometry smoke

Automatic run provenance

Field	Recorded value
Run ID	20261026T184156302988Z-0fd25519
Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	5b44233833e827508849902ce91a586195982be36062c140f55ae5740d76b7a8
Portable protocol diagnostic SHA-256	88551245401d393227ab683122d09f1c9b2360873f7a74af6b6cea24d4ef9490
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	f5dfada0e1b16b915daeb18327c7b9c2306d4583960e7192152fd38f1a0a3aa1
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason

G0	PASS	Are the qualified framework and dedicated adapter protocol byte-exact?	Qualified framework and dedicated adapter protocol verified; not the full framework blind/reveal FREEZE.
G1	PASS	Do all selected geometry sources satisfy exact admission and hashes?	Rehashed exact admitted geometry and prior capability evidence; geometry authority only.
G2	PASS	Are inherited geometry metrics finite under the fixed bounded ladder?	Inherited geometry definitions, native units and registered smoke budget; no physical closure.
G3	NOT_RUN_PREREQUISITE	Are independent physical discovery inputs available and qualified?	Required scientific inputs unavailable: Independent physical energies, gaps, relaxation/couplings, sampled state counts, force/ pressure holdouts and calibrated SI scale are not supplied by the admitted geometry atlas.
G4	NOT_RUN_PREREQUISITE	Does independent physical held-out evidence confirm a mechanism?	Required scientific inputs unavailable: Independent physical energies, gaps, relaxation/couplings, sampled state counts, force/ pressure holdouts and calibrated SI scale are not supplied by the admitted geometry atlas.
G5	PASS	Does the inherited native geometry kernel match reference tolerances?	Measured despite unavailable physical prerequisites; parity never certifies physical SST.
G6	NOT_APPLICABLE	GPU screening (disabled in this CPU geometry tranche)	Disabled by framework profile
G7	NOT_RUN_PREREQUISITE	Is independent physical replication available?	Required scientific inputs unavailable: Independent physical energies, gaps, relaxation/couplings, sampled state counts, force/ pressure holdouts and calibrated SI scale are not supplied by the admitted geometry atlas.
G8	NOT_RUN_PREREQUISITE	Is physical mechanism and convergence evidence available?	Required scientific inputs unavailable: Independent physical energies, gaps, relaxation/couplings, sampled state counts, force/ pressure holdouts and calibrated SI scale are not supplied by the admitted geometry atlas.
G9	DEFERRED	Reveal is separate and not allowed in this geometry-only tranche	No reveal requested or permitted by this geometry-only tranche.

Scientific qualification (separate from measured gate outcomes)

Gate	Evidence qualification	Blocking prerequisites
G0	QUALIFIED	none
G1	QUALIFIED	none
G2	QUALIFIED	none
G3	BLOCKED_PREREQUISITE	none
G4	BLOCKED_PREREQUISITE	G3
G5	DIAGNOSTIC_ONLY	G4
G6	NOT_APPLICABLE	G5
G7	BLOCKED_PREREQUISITE	G6
G8	BLOCKED_PREREQUISITE	G7
G9	DEFERRED	G8

Backend manifest

{

```
  "actual_backend": {
    "backend": "cpp",
    "native": true,
    "version": "0.3.1-cpp17"
  },
  "authority": "GEOMETRY_KERNEL_ONLY",
  "gpu_used": false,
  "native_binary_sha256":
"9c387369427ff986101d4f345ac75d2a24d8ec640f24999132b3fc351f5294f6",
  "native_cpp_sha256":
"00bd3ae13925fb176b501b676c89e4a95c72b6ac0cea6fd297b0b0b990dcf0cd",
  "native_rebuilt": false,
  "numpy": "2.4.4",
  "physical_qualification": "UNRESOLVED",
  "python": "3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944
 64 bit (AMD64)]",
  "schema": "SST-BACKEND-MANIFEST-2"
}
```


Source manifest


```
{
  "manifest_sha256":
"616420b8dbd5d89777e7bfd9558518b5805a6c91bdb33f64e2019cc718bd516f",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "knotplot_relaxation",
      "path": "C:\\workspace\\projects\\SST-Workbench\\03_data\\A_knots\\04_k
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_rr_010k_coarse.atstart.vect",
      "provenance_family": null,
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
      },
      "role": "admitted_geometry_diagnostic",
      "sha256":
"54d6e30010753bb15e4a47f84ec874a0420bcd3bc5e5264edd1a54b508467c3f",
      "source_id": "CAR_17fca17caa5ca443",
      "version": "E010-v0.3.1/view-v0.3.4"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
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      "path": "C:\\workspace\\projects\\SST-Workbench\\03_data\\A_knots\\04_k
notplot\\knot_3.1\\knot_3.1_trial_002k_rr_010k_coarse_rr_050k_eqfinal_rr_030k
_polish.vect",
      "provenance_family": null,
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
      },
      "role": "admitted_geometry_diagnostic",
      "sha256":
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      "source_id": "CAR_042c4a4f554aef25",
      "version": "E010-v0.3.1/view-v0.3.4"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "knotplot_relaxation",
      "path": "C:\\workspace\\projects\\SST-Workbench\\03_data\\A_knots\\04_k
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t_4.1_trial_001k_rr_010k_coarse_rr_050k_eqfinal.vect",
      "provenance_family": null,
      "registry_hint": {
        "catalog_id": "E010",
```

```
        "hits": []
      },
      "role": "admitted_geometry_diagnostic",
      "sha256":
"50b548c10790e94457bfe8e414ba9e7f144ff6b12cea46e63f870eb9d460a333",
      "source_id": "CAR_0b173a215edeb25c",
      "version": "E010-v0.3.1/view-v0.3.4"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation, falsification limits and unresolved claims

A finite geometry metric or kernel-parity PASS supports only the named implementation diagnostic. It does not establish an energy spectrum, thermodynamic closure, gravity, Euler/Casimir mechanism or new SST prediction. A parity or numerical failure is a failed software check, not independent physical falsification. Missing source calibration, held-out fields, energies, gaps, state counts and convergence evidence remain explicit. Historical negatives and prior runs are preserved.

16 Reproducibility and execution instructions

From this version directory on Windows execute `run_all.cmd`; `run_all.cmd SELFTEST` runs tests only. CPython 3.14, the pinned native binary and the qualified local framework/source paths are required. Scientific PDF generation uses the framework LaTeX compiler and fails closed if it cannot compile. Outputs use the declared catalog/family prefix and `_outputs`; prior canonical output is retained under its run ID. PDF and ZIP publications are one directory above this version and include immutable run IDs. The adjacent SHA-256 verifies each ZIP. `LATEST_RUN.json` is the only replaceable publication pointer; `PUBLICATIONS.jsonl` is append-only.

17 References and primary reproducibility records

1. `science_contract.json`, original registered hypotheses, symbols, equations E1/E2 and code-to-equation steps.
2. `src/maxwell_sst_falsifier/geometry.py`, exact inherited geometric definitions; `native_ext/fallback.py` and `cpp/native.cpp`, reference/native kernels.
3. `provenance/A011_DATASET_ADMISSION.json` and carrier decision ledger; exact E010 parser code pins and raw/geometry source hashes.
4. `preregistration/FROZEN_PROTOCOL.json` and `FRAMEWORK_SOURCE_PIN.json`: byte-exact geometry protocol and qualified v1.0.10 source.
5. Current `GATE_LEDGER.json`, `REPORT_DATA.json`, `SCIENCE_EVIDENCE_MANIFEST.json` and final `OUTPUT_MANIFEST.json`.